

WHERE Conditions

Learning Objectives

- Use comparison operators to filter records in SQL queries
- Apply logical operators AND, OR, and NOT to combine conditions
- Use BETWEEN, IN, and LIKE operators for advanced filtering

Engage (5-7 minutes)

A librarian wants to find all books published after 2020 that cost less than 500 rupees. Combining two conditions narrows the search. The WHERE clause with logical operators makes this filtering possible.

Explore (10-12 minutes)

Given a Products table, write queries using: WHERE Price > 100, WHERE Price BETWEEN 100 AND 500, WHERE Category IN ('Electronics', 'Books'), and WHERE Name LIKE 'A%'. Test each and observe how results change based on the condition used.

Explain (10-12 minutes)

Comparison operators include =, !=, <, <=, >, >=. Logical operators combine conditions: AND requires both true, OR requires either true, NOT negates a condition. BETWEEN selects within a range, IN matches a list of values, and LIKE pattern-matches text using % for any characters and _ for single characters.

Elaborate (8-10 minutes)

A school wants to find students who scored above 80 in Science AND above 75 in Maths, or students who scored above 90 in English. Write this complex query using AND, OR, and parentheses to ensure correct evaluation order.

Evaluate (5-8 minutes)

Write WHERE clauses for: students enrolled after January 2024, products priced between 50 and 200, names starting with 'S', and categories that are NOT 'Clearance'. Explain operator precedence in AND and OR combinations.